AEX-TFANET : Adaptive Explainable Trust Framework for Secure Flying Ad Hoc Networks
PID No.- 23CSEP17

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Problem Identification

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Problem Identification

- Flying Ad Hoc Networks (FANETs) enable UAVs to communicate without relying completely on fixed infrastructure.
- UAVs operate with **high mobility and rapidly changing network topology**.
- Dynamic links make reliable routing difficult.
- FANETs are vulnerable to malicious and selfish UAVs.
- Attacks such as **Sybil, blackhole, packet dropping, spoofing and denial-of-service** can disrupt communication.
- Traditional routing protocols generally do not consider the continuously changing trustworthiness of UAV nodes.
- AI-based security systems may provide decisions without clearly explaining why a UAV or route was considered unsafe.

Need:
Adaptive Trust + Secure Routing + Explainable Decision Making

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Problem and Key Challenges

Network Challenges

- Highly dynamic UAV topology
- Frequent link failures
- Rapid neighbour changes
- Limited communication range
- Resource and energy constraints

Security Challenges

- Malicious UAVs
- Sybil attacks
- Blackhole and packet dropping
- False information
- Denial-of-Service attacks
- Compromised insider nodes

Core Problem:
Develop an adaptive framework that evaluates UAV trustworthiness, selects secure communication paths, detects malicious behaviour, and explains its security decisions.

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Objectives

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Objectives

- Develop an **adaptive trust management framework** for FANETs.
- Continuously evaluate the trustworthiness of UAV nodes.
- Consider packet forwarding, communication reliability, link stability and historical behaviour.
- Detect potentially malicious or unreliable UAVs.
- Use trust values during secure route selection.
- Adapt routing decisions according to changing network conditions.
- Incorporate **Explainable AI (XAI)** for transparent security decisions.
- Reduce packet loss and unreliable routes.
- Improve network security, reliability and resilience.
- Evaluate the proposed framework using FANET simulation.

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Literature Review

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Literature Review

- **Dey et al. (2024):** Reviewed trust-based mechanisms for FANETs and highlighted trust computation for identifying untrustworthy UAVs.
- **Diockou et al. (2024):** Reviewed FANET security mechanisms including secure routing, intrusion detection, lightweight cryptography and trust management.
- **Fuzzy Trust-Based Routing (2023):** Used local and path trust mechanisms to identify reliable neighbours and reduce malicious route formation.
- **Alkahtani et al. (2026):** Proposed an explainable multi-agent reinforcement learning framework combining trust-based security with XAI techniques.
- Recent research shows increasing interest in combining **trust, adaptive learning and explainability** for secure UAV communication.

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Literature Comparison

Approach	Advantages	Limitations
Traditional Routing	Simple and widely used	Does not consider node trust
Cryptographic Security	Authentication and confidentiality	Computational overhead
Trust-Based Routing	Detects unreliable nodes	Trust calculation can be complex
Fuzzy Trust	Handles uncertain behaviour	Depends on suitable rules
AI-Based Security	Adaptive decision making	Often difficult to interpret
XAI-Based Security	Provides understandable decisions	Additional computation
AEX-TFANET	Trust + Adaptation + XAI	Requires simulation and tuning

Observation:
Existing approaches address individual security or routing problems, but there is a need for a unified framework combining **adaptive trust, secure routing and explainable AI**.

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Research Gap

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Research Gap

- Many FANET security approaches use static or predefined trust parameters.
- Rapidly changing UAV behaviour requires **continuous trust adaptation**.
- Traditional trust mechanisms may not adequately explain why a node is classified as malicious.
- AI-based routing can improve adaptability but may behave as a black box.
- Security decisions need to be interpretable for network operators.
- Trust information should influence routing dynamically.
- Existing approaches often evaluate security and routing performance separately.
- A unified approach combining **trust evaluation, adaptive routing and explainability** remains an important research direction.

Research Gap:

Trust + Adaptation + Secure Routing + Explainability

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Methodology

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Proposed Methodology

1. **FANET Initialization**
   - Deploy multiple UAV nodes.
   - Establish communication links between neighbours.
2. **Network Monitoring**
   - Collect node and communication behaviour.
   - Monitor packet forwarding and link characteristics.
3. **Trust Calculation**
   - Calculate direct and indirect trust values.
   - Combine behavioural and network parameters.
4. **Adaptive Trust Engine**
   - Continuously update trust based on observations.
5. **Secure Route Selection**
   - Prefer highly trusted and reliable UAVs.
6. **Threat Detection**
   - Identify abnormal or malicious behaviour.
7. **Explainability Layer**
   - Explain factors influencing trust and routing decisions.
8. **Performance Evaluation**
   - Measure PDR, delay, throughput and packet loss.

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Adaptive Trust Classification

Trust Levels

- **Highly Trusted**
  - Consistent packet forwarding
  - Stable communication
- **Trusted**
  - Generally reliable behaviour
- **Suspicious**
  - Unusual packet behaviour
  - Decreasing reliability
- **Malicious / Untrusted**
  - Repeated packet dropping
  - Abnormal behaviour

Trust Factors

- Packet forwarding ratio
- Packet delivery behaviour
- Link stability
- Communication consistency
- Behaviour history
- Neighbour recommendations
- Anomaly indicators

Output

- Trust Score
- Route Decision
- Security Action

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System Architecture

UAV Nodes

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FANET Communication

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Network Monitoring

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Trust Data Collection

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Adaptive Trust Engine

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Malicious Node Detection

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Secure Route Selection

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Explainable AI Layer

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Security Decision

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Performance Monitoring

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Technology Stack

Simulation / Network

- NS-3
- FANET Mobility Models
- UAV Network Simulation
- Network Performance Analysis

AI / Trust Management

- Python
- Machine Learning
- Trust Scoring
- Anomaly Detection

Explainable AI

- SHAP
- LIME
- Feature Contribution Analysis

Data / Analysis

- NumPy
- Pandas
- Matplotlib
- CSV Simulation Results

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Work Plan

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Work Plan

Phase	Activities	Status
Problem Identification	Define FANET security problem	Completed
Literature Review	Trust, routing and XAI research	Completed
FANET Design	Define UAV network model	Planned
Trust Model	Develop trust calculation mechanism	Planned
Adaptive Engine	Dynamic trust updating	Planned
Secure Routing	Trust-aware route selection	Planned
XAI Module	Explain trust and routing decisions	Planned
Simulation	NS-3-based experiments	Planned
Testing	Attack and performance evaluation	Planned
Final Integration	Complete AEX-TFANET framework	Planned

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Expected Outcome

- Adaptive trust evaluation of UAV nodes.
- Improved identification of malicious or unreliable nodes.
- Secure route selection based on trust.
- Better resilience against malicious behaviour.
- Reduced packet loss caused by unreliable nodes.
- Improved packet delivery ratio.
- Adaptive response to changing network conditions.
- Explainable security and routing decisions.
- Better transparency of AI-based decisions.
- Simulation-based evaluation of the proposed AEX-TFANET framework.

Proposed System:

Adaptive + Secure + Trust-Aware + Explainable

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References

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